

SATHISH R

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Education

Rajalakshmi Engineering College, Chennai

B. Tech in Information Technology

Chennai, Tamil Nadu, India

Sep 2024 – Present

CGPA : 8.02

Summary

Backend Engineering undergraduate with hands-on experience developing scalable applications using Java, Spring Boot, MongoDB, and REST APIs. Built full-stack and machine learning projects involving authentication, deployment, and predictive analytics. Passionate about solving real-world problems through efficient backend systems and continuously improving software engineering skills.

Technical Skills

Programming : Java, Python , C

Web Development : HTML, CSS, React, Spring Boot

Database Management : MySQL, MongoDB

Tools : VS Code, Jupyter Notebook, Git, GitHub

Concepts : Object Oriented Programming, Data Structures, Machine Learning

Projects

Developer Knowledge Vault

React, Spring Boot, MongoDB

- Developer notes and coding references were scattered across multiple platforms with no centralized system for retrieval or organization.
- Designed and built a full-stack knowledge management platform to centralize, search, and version-control developer notes.
- Engineered Spring Boot 3.2.5 backend with 17 REST endpoints, JWT authentication, MongoDB Text Indexing, hybrid React-side filtering, and automatic version history snapshots.
- Delivered a secure multi-user platform with instant zero-latency search, access analytics dashboard, and full revision history for every note.

Crop & Carbon Prediction System

Python, FastAPI, Scikit-learn, Docker

- Farmers lacked accessible data-driven tools to determine suitable crops based on soil and climate conditions.
- Built an end-to-end ML web application predicting optimal crops and soil carbon sequestration potential.
- Trained Random Forest Classifier and Regressor on 7 parameters (N, P, K, pH, rainfall, temperature, humidity), served via FastAPI, containerized with Docker.
- Deployed live on Render cloud with real-time dual-model predictions and a publicly accessible web interface.
- Website link : <https://crop-and-carbon-score-prediction.onrender.com>

Achievements

- Solved 90+ Data Structures and Algorithms problems on LeetCode, covering arrays, linked lists, stacks, queues and recursion.
- Earned the 50 Days LeetCode Badge through consistent daily problem-solving.

Certifications

- Artificial Intelligence & Machine Learning – IITM Pravartak
- Software Engineering Virtual Experience – JPMorgan Chase & Co. (Forage).